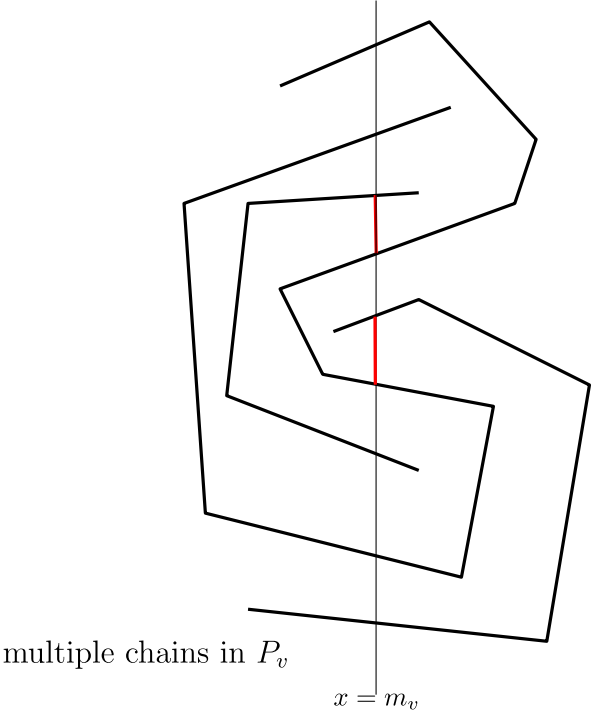
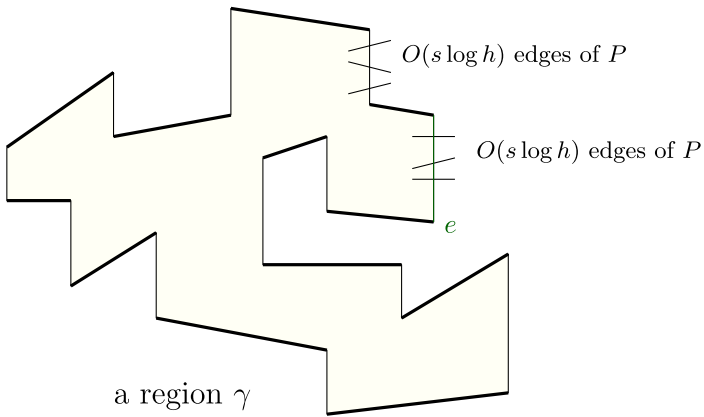

multiple chains in P_v
$x = m_v$


$O(s \log h)$ edges of P
$O(s \log h)$ edges of P
e
a region γ